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StudyGroupAsker

StudyGroup ask()
StudyGroup ask(Int)

CoordinatesAsker

Coordinates ask()

Asker

T ask()

SemesterAsker

Semester? ask()

PersonAsker

Person ask()

Show

ExitCode execute()

Clear

ExitCode execute()

Exit

ExitCode execute()

Undo

ExitCode execute()

RemoveById

ExitCode execute()

CountGreaterThanAverageMark

ExitCode execute()

Help

ExitCode execute()

RemoveLast

ExitCode execute()

ExecuteScript$Companion

MutableSet<String> activeScripts

ExecuteScript

ExitCode execute()

InsertAt

ExitCode execute()

CountLessThanAverageMark

ExitCode execute()

AddIfMax

ExitCode execute()

Info

ExitCode execute()

Add

ExitCode execute()

Command

ExitCode execute()

Update

ExitCode execute()

AverageOfAverageMark

ExitCode execute()

RUDPReceiver

IResponse receiveResponse()

RUDPClient

IResponse sendAndReceive(IRequest)

RUDPSender

void ping()
void sendPackets(List<RUDPPacket>)
void sendACK()

IRUDPClient

IResponse sendAndReceive(IRequest)

IConsole

Boolean fileMode
BufferedReader reader

String readLine()
void printLine(String)
void printObject(Any)
void printError(Exception)
void printError(String)
void setFileMode(String)
void setInteractiveMode()

Console

String readLine()
void printLine(String)
void printObject(Any)
void printError(Exception)
void printError(String)
void setFileMode(String)
void setInteractiveMode()

CommandManager

void register(Command)
Command getCommand(String)

ICommandManager

MutableMap<String, Command> commands

void register(Command)
Command getCommand(String)

PrettyMap

String toString()

RUDPServerSender

void sendResponse(List<RUDPPacket>,InetAddress,Int)

RUDPServer

void runner()

RUDPServerReceiver

Pair<InetAddress, Int> receivePing(DatagramPacket)
IRequest receiveRequest(InetAddress,Int)

HealthCheckListener

void run()

Runnable

AddCommand

isUndoable

IResponse execute(IRequest)
IResponse undo()

ServerCommand

Boolean isUndoable
Boolean isMutating

IResponse execute(IRequest)
IResponse undo()

RemoveByIdCommand

isUndoable

IResponse execute(IRequest)
IResponse undo()

ClearCommand

isUndoable

IResponse execute(IRequest)
IResponse undo()

CountLessThanAverageMarkCommand

IResponse execute(IRequest)

AverageOfAverageMarkCommand

IResponse execute(IRequest)

RemoveLastCommand

isUndoable

IResponse execute(IRequest)
IResponse undo()

CountGreaterThanAverageMarkCommand

IResponse execute(IRequest)

ShowCommand

IResponse execute(IRequest)

InsertAtCommand

isUndoable

IResponse execute(IRequest)
IResponse undo()

InfoCommand

IResponse execute(IRequest)

SaveCommand

IResponse execute(IRequest)

UndoCommand

isM utating

IResponse execute(IRequest)

UpdateCommand

isUndoable

IResponse execute(IRequest)
IResponse undo()

AddIfMaxCommand

isUndoable

IResponse execute(IRequest)
IResponse undo()

IFileManager

String fileName

String writeCollection(Collection<StudyGroup>)
Pair<String, List<StudyGroup>>? readCollection()
void clearVersion(Collection<StudyGroup>)

ICollectionManager

Vector<StudyGroup> collection
LocalDateTime lastInitTime
LocalDateTime lastEditTime
String currentVersion

void add(StudyGroup)
void updateById(Int,StudyGroup)
void removeById(Int)
void clear()
void insertAt(Int,StudyGroup)
void removeLast()
Boolean addIfMax(StudyGroup)
Long getAverageMarkFromAll()
Long countAverageMarkLessThen(Long)
Long countAverageMarkGreaterThen(Long)
List<StudyGroup> takeSnapshot()
void restoreSnapshot(List<StudyGroup>)

IRequestManager

IResponse dispatch(IRequest)

FileWatcher

void run()

FileManager

String writeCollection(Collection<StudyGroup>)
Pair<String, List<StudyGroup>>? readCollection()
void clearVersion(Collection<StudyGroup>)

RequestManager

IResponse dispatch(IRequest)
void saveCollection()
void clearVersionOnShutdown()

CollectionManager

void add(StudyGroup)
void updateById(Int,StudyGroup)
void removeById(Int)
void clear()
void insertAt(Int,StudyGroup)
void removeLast()
Boolean addIfMax(StudyGroup)
Long getAverageMarkFromAll()
Long countAverageMarkLessThen(Long)
Long countAverageMarkGreaterThen(Long)
List<StudyGroup> takeSnapshot()
void restoreSnapshot(List<StudyGroup>)

Runnable

RUDPBalancer

void start()

ServerAddress

BalancerConfig n u ll

RequestForwarder

void sendPing(InetAddress,Int)
void sendRequest(List<RUDPPacket>,InetAddress,Int)

ServerResponseReceiver

IResponse receiveResponse(InetAddress,Int)

ResponseForwarder

void sendResponse(List<RUDPPacket>,InetAddress,Int)

NodeSelector

ServerNode select()

ServerNode

address

Boolean isAvailable()
void markDead()
void markAlive()

NodeHealthChecker

void run()

Runnable

BalancerReceiver

Pair<InetAddress, Int> receivePing(DatagramPacket)
IRequest receiveRequest(InetAddress,Int)

StudyGroupFactory

Int nextId()
void syncIdGenerator(Int)
void decrementId()
StudyGroup assignServerFields(StudyGroup)
StudyGroup forUpdate(StudyGroup,Int,LocalDateTime)

StudyGroupParser

StudyGroup parse(LineReader)
StudyGroup parse(LineReader,Int)

LineReader

String readLine()
Boolean hasNext()

Assembler$Companion

Assembler

void addPacket(RUDPPacket)
Boolean isComplete(Uuid)
ByteArray assemble(Uuid)

IAssembler

Map<Uuid, List<RUDPPacket>> map

void addPacket(RUDPPacket)
Boolean isComplete(Uuid)
ByteArray assemble(Uuid)

AddRequest InsertAtRequest CountLessThanAverageMarkRequest AverageOfAverageMarkRequest

IRequest

CommandName commandName

AddResponse CommandResponse RemoveByIdRequest ExecuteScriptRequest ShowRequest UndoRequest UpdateRequest RemoveLastRequest AddIfMaxRequest ClearRequest InfoRequest CountGreaterThanAverageMarkRequest

IResponse

ExitCode exitCode

RUDPPacket$Companion

HEADING

RUDPPacket fromByteBuffer(ByteBuffer)
RUDPPacket fromByteArray(ByteArray)
ByteBuffer byteBufferACK()
ByteBuffer byteBufferPING()
ByteBuffer byteBufferFIN()
ByteArray byteArrayACK()
ByteArray byteArrayPING()
ByteArray byteArrayFIN()
Boolean isByteBufferACK(ByteBuffer)
Boolean isByteBufferPING(ByteBuffer)
Boolean isByteBufferFIN(ByteBuffer)
Boolean isByteArrayACK(ByteArray)
Boolean isByteArrayPING(ByteArray)
Boolean isByteArrayFIN(ByteArray)

RUDPPacket

ByteBuffer toByteBuffer()
ByteArray toByteArray()

CommandName

HELP
INFO
SHOW
ADD
UPDATE
REMOVE_BY_ID
CLEAR
EXECUTE_SCRIPT
EXIT

INSERT_AT
REMOVE_LAST
ADD_IF_MAX
AVERAGE_OF_AVERAGE_MARK
COUNT_LESS_THAN_AVERAGE_MARK
COUNT_GREATER_THAN_AVERAGE_MARK
UNDO

ExitCode

OK
ERROR
EXIT

DateSerializer

SerialDescriptor descriptor

void serialize(Encoder,Date)
Date deserialize(Decoder)

StudyGroup

Int compareTo(StudyGroup)

Coordinates

Semester

THIRD
SEVENTH
EIGHTH

Person

Comparable

ValidationException UndoException ScriptErrorException ScriptRecursionException CommandNotFoundException NotFoundException ServerTimeoutException

Exception

KotlinJvmPlugin

void apply(org.gradle.api.Project)

Plugin

LibrariesForLibs

Provider<MinimalExternalModuleDependency> getKotlinGradlePlugin()
Provider<MinimalExternalModuleDependency> getKotlinxCoroutines()
Provider<MinimalExternalModuleDependency> getKotlinxDatetime()
Provider<MinimalExternalModuleDependency> getKotlinxSerialization()
VersionAccessors getVersions()
BundleAccessors getBundles()
PluginAccessors getPlugins()

LibrariesForLibs$VersionAccessors

Provider<String> getKotlin()
Provider<String> getKotlinxCoroutines()
Provider<String> getKotlinxDatetim e()
Provider<String> getKotlinxSerializationJSON()

LibrariesForLibs$BundleAccessors

Provider<ExternalModuleDependencyBundle> getKotlinxEcosystem()

LibrariesForLibs$PluginAccessors

Provider<PluginDependency> getKotlinPluginSerialization()

LibrariesForLibsInPluginsBlock

Provider<MinimalExternalModuleDependency> getKotlinGradlePlugin()
Provider<MinimalExternalModuleDependency> getKotlinxCoroutines()
Provider<MinimalExternalModuleDependency> getKotlinxDatetime()
Provider<MinimalExternalModuleDependency> getKotlinxSerialization()
VersionAccessors getVersions()
BundleAccessors getBundles()
PluginAccessors getPlugins()

LibrariesForLibsInPluginsBlock$VersionAccessors

Provider<String> getKotlin()
Provider<String> getKotlinxCoroutines()
Provider<String> getKotlinxDatetim e()
Provider<String> getKotlinxSerializationJSON()

LibrariesForLibsInPluginsBlock$BundleAccessors

Provider<ExternalModuleDependencyBundle> getKotlinxEcosystem()

LibrariesForLibsInPluginsBlock$PluginAccessors

Provider<PluginDependency> getKotlinPluginSerialization()

PluginFactory BundleFactory VersionFactory

OrgPluginGroup OrgGradlePluginGroup OrgJetbrainsPluginGroup AbstractExternalDependencyFactory

OrgJetbrainsKotlinPluginGroup OrgJetbrainsKotlinNativePluginGroup OrgJetbrainsKotlinPluginPluginGroup

KSerializer

MockRUDPClient

net.requests.IRequest? lastRequest

IResponse sendAndReceive(net.requests.IRequest)

CommandsTest

void setUp()
void add_should_send_add_request_and_return_ok()
void show_should_send_show_request_and_return_ok()
void clear_should_send_clear_request_and_return_ok()
void info_should_send_info_request_and_return_ok()
void help_should_return_ok_and_list_commands()
void exit_should_return_exit_code()
void remove_last_should_send_remove_last_request()
void remove_by_id_should_send_remove_by_id_request()
void undo_should_send_undo_request()
void average_of_average_mark_should_send_request()
void count_less_than_average_mark_should_send_request()
void count_greater_than_average_mark_should_send_request()
void command_should_return_error_when_server_returns_error()

TestConsole

lines

String readLine()
void printLine(String)
void printObject(Any)
void printError(Exception)
void printError(String)
void setFileMode(String)
void setInteractiveMode()

ExecuteScriptTest$MockClient

net.requests.IRequest? lastRequest

IResponse sendAndReceive(net.requests.IRequest)

ExecuteScriptTest

void executeScript_sends_raw_lines_to_server()
void executeScript_expands_nested_script_inline()
void executeScript_detects_self_referencing_recursion()
void executeScript_detects_indirect_recursion()
void executeScript_returns_error_for_nonexistent_file()
void executeScript_returns_ok_for_empty_file()
void executeScript_propagates_server_exit_code()

NoOpFileManager

fileNam e

void writeCollection(Collection<StudyGroup>)
List<StudyGroup>? readCollection()

ServerCommandsTest

void setUp()
void add_increases_collection_size()
void add_element_is_present_in_collection()
void add_if_max_adds_element_when_it_exceeds_all_existing()
void add_if_max_does_not_add_when_element_is_not_max()
void add_if_max_adds_to_empty_collection()
void show_returns_ok_for_empty_collection()
void show_returns_all_elements_when_collection_is_not_empty()
void info_returns_collection_type_and_size()
void clear_empties_a_non_empty_collection()
void clear_on_empty_collection_returns_ok()
void remove_last_removes_the_last_element()
void remove_last_on_empty_collection_returns_error()
void remove_by_id_removes_the_correct_element()
void remove_by_id_returns_error_for_nonexistent_id()
void insert_at_places_element_at_given_index()
void update_replaces_element_with_matching_id()
void update_returns_error_for_nonexistent_id()
void average_of_average_mark_returns_zero_for_empty_collection()
void average_of_average_mark_calculates_mean_correctly()
void count_less_than_average_mark_counts_correctly()
void count_less_than_average_mark_returns_zero_when_none_qualify()
void count_greater_than_average_mark_counts_correctly()
void undo_reverts_last_add()
void undo_reverts_last_clear()
void undo_reverts_multiple_commands_in_order()
void undo_on_empty_history_returns_error()
void undo_reverts_remove_by_id()
void undo_reverts_update()
void execute_script_parses_and_executes_add_lines()
void execute_script_rolls_back_all_changes_when_any_command_fails()
void execute_script_rollback_restores_history()
void execute_script_returns_ok_for_empty_lines()
void execute_script_skips_unknown_command_names()
void execute_script_accumulates_output_from_all_commands()
void execute_script_parse_error_rolls_back()
void dispatch_returns_ok_for_all_known_read_only_commands()
void collection_manager_snapshot_preserves_state()
void add_then_update_then_undo_restores_original()
void clear_then_undo_then_remove_last_leaves_empty_collection()

SerializationTest

void studyGroup_survives_protobuf_roundtrip()
void addRequest_roundtrip_preserves_study_group()
void addIfMaxRequest_roundtrip()
void showRequest_roundtrip()
void infoRequest_roundtrip()
void clearRequest_roundtrip()
void removeLastRequest_roundtrip()
void removeByIdRequest_roundtrip_preserves_id()
void insertAtRequest_roundtrip_preserves_index_and_group()
void updateRequest_roundtrip_preserves_id_and_group()
void averageOfAverageMarkRequest_roundtrip()
void countLessThanAverageMarkRequest_roundtrip_preserves_threshold()
void countGreaterThanAverageMarkRequest_roundtrip_preserves_threshold()
void undoRequest_roundtrip_preserves_n()
void executeScriptRequest_roundtrip_preserves_lines()
void executeScriptRequest_empty_lines_roundtrip()
void commandResponse_ok_roundtrip()
void commandResponse_error_roundtrip()
void commandResponse_empty_message_roundtrip()
void polymorphic_sealed_dispatch_works_for_all_request_types()